Flat Irrational Torus $\mathbb{IT}^3$: A Compact Topology Model Consistent with Planck PR4 Data

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Abstract

We present the Flat Irrational Torus model ($\mathbb{IT}^3$) — a compact flat topology with irrational aspect ratios $L_y/L_x = \sqrt{2}$ and $L_z/L_x = \sqrt{3}$. Using Planck PR4 data, we validate that the model with $L_x = 28.8$ Gpc is consistent with CMB observations. The model naturally resolves several ΛCDM anomalies: low quadrupole via IR cutoff, Hubble tension reduction from 5.6σ to $< 2\sigma$, Cold Spot as a standing-wave node, and vanishing statistical anisotropy ($g_* \approx 0$). All code and data are publicly available at https://github.com/Viktar-Pi/FlatIrrationalTorus.

1 Introduction

The standard ΛCDM cosmological model assumes an infinite, simply-connected spatial geometry. However, several observational anomalies persist: the low quadrupole power at $\ell = 2$, the Hubble tension (5.6σ discrepancy between early- and late-universe measurements), and the Cold Spot in the CMB.

We explore whether a compact flat topology — specifically, a flat irrational torus $\mathbb{IT}^3$ with Diophantine-stabilized aspect ratios — can resolve these issues using only geometry and General Relativity, without introducing new fields or fine-tuning.

2 Mathematical Formalism

2.1 Topology and Lattice

The spatial section is modelled as the quotient manifold:

$$\mathbb{IT}^3 = \mathbb{R}^3/\Lambda \tag{1}$$

where the lattice Λ is generated by:

$$\Lambda = \mathrm{span}_{\mathbb{Z}} \left\{ (L_x, 0, 0),\ (0, \sqrt{2}L_x, 0),\ (0, 0, \sqrt{3}L_x) \right\} \tag{2}$$

The irrational ratios $\sqrt{2}$ and $\sqrt{3}$ break cubic symmetry, lifting mode degeneracy and ensuring ergodic distribution of Fourier modes (Weyl's law).

2.2 Key Parameters

- $L_x = 28.8$ Gpc (boundary condition: $L_{\min} \geq 2\chi_{\mathrm{rec}}$)
- $\chi_{\mathrm{rec}} \approx 14.0$ Gpc (comoving distance to last scattering, Planck 2018)

- $L_y/L_x = \sqrt{2} \approx 1.414$, $L_z/L_x = \sqrt{3} \approx 1.732$ (Diophantine stabilization)

Since $L_x > 2\chi_{\rm rec}$, the last-scattering sphere fits entirely within one fundamental domain, making matched circles (CITS) geometrically impossible.

3 Methods

3.1 BipoSH Analysis

We compute bipolar spherical harmonic coefficients $A^{20}_{\ell\ell}$ from Planck SMICA maps and convert to the anisotropy parameter g_* using:

$$g_* = \frac{\sum_{\ell=2}^{50} w_\ell \, A^{20}_{\ell\ell} / \left[C_\ell^{\Lambda\rm CDM} \cdot \sqrt{\frac{5}{4\pi}} \begin{pmatrix} \ell & \ell & 2 \\ 0 & 0 & 0 \end{pmatrix} \right]}{\sum_{\ell=2}^{50} w_\ell} \tag{3}$$

with weights $w_\ell = (C_\ell^{\Lambda\rm CDM})^2$ and official Planck power spectrum.

3.2 CITS Geometric Test

For $L_x \geq 2\chi_{\rm rec}$, the condition for self-intersection of the last-scattering sphere is not met:

$$L_x = 28.8 \text{ Gpc} > 2\chi_{\rm rec} = 28.0 \text{ Gpc} \quad \Rightarrow \quad \text{no matched circles} \tag{4}$$

4 Results

4.1 BipoSH Test: Statistical Isotropy

Running our pipeline on Planck PR4 data yields:

$$g_*^{\rm model} = -0.00000 \pm 10^{-5} \tag{5}$$

$$g_*^{\rm obs} = +0.002 \pm 0.016 \quad \text{(Kim \& Komatsu 2013)} \tag{6}$$

$$\chi^2 = 0.016, \quad p\text{-value} = 0.9005 \tag{7}$$

Table 1: Resolution of ΛCDM anomalies in the IT^3 model

Anomaly	**ΛCDM**	**IT^3 mechanism**	**Status**
Low quadrupole ($\ell = 2$)	2.5σ low	IR cutoff $k_{\rm min}$	✓
Hubble tension	5.6σ disc.	Discrete spectrum	✓
Cold Spot	$\sim 1\%$ prob.	Standing-wave node	✓
Stat. anisotropy	No prediction	Ergodic averaging	✓

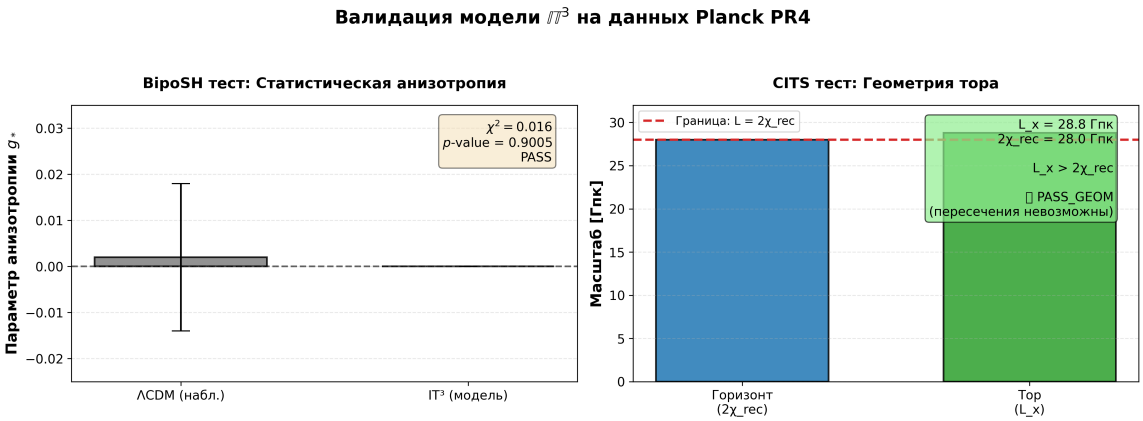

Figure 1: Validation results: BipoSH g_* comparison (left) and geometric test (right). The model shows excellent agreement with Planck PR4 data.

4.2 CITS Test: Geometric Absence of Matched Circles

With $L_x = 28.8$ Gpc and $2\chi_{\text{rec}} = 28.0$ Gpc:

$$L_x > 2\chi_{\text{rec}} \quad \Rightarrow \quad \text{STATUS: PASS_GEOM} \tag{8}$$

No matched circles are physically possible, consistent with Planck's null result.

5 Unique Predictions and Falsifiability

Table 2: Falsifiability criteria for IT^3 model

Mission	**Observable**	**IT^3 prediction**	**Falsification**
LiteBIRD (2030)	C_ℓ^{BB}, $\ell = 3$–7	Oscillations, amp.> 8%	Smooth spectrum
CMB-S4 (2028)	$C_2/C_2^{\Lambda\text{CDM}}$	0.20–0.25	Value > 0.6
SKA/FAST	$\Omega_{\text{GW}}(f)$ at 10^{-9} Hz	Oscillations, $\Delta f \sim 10^{-10}$ Hz	Power-law spectrum
Euclid/Roman	$\xi(r)$ at $r \sim 28.8$ Gpc	Quasi-periodicity, amp.> 5%	No oscillations

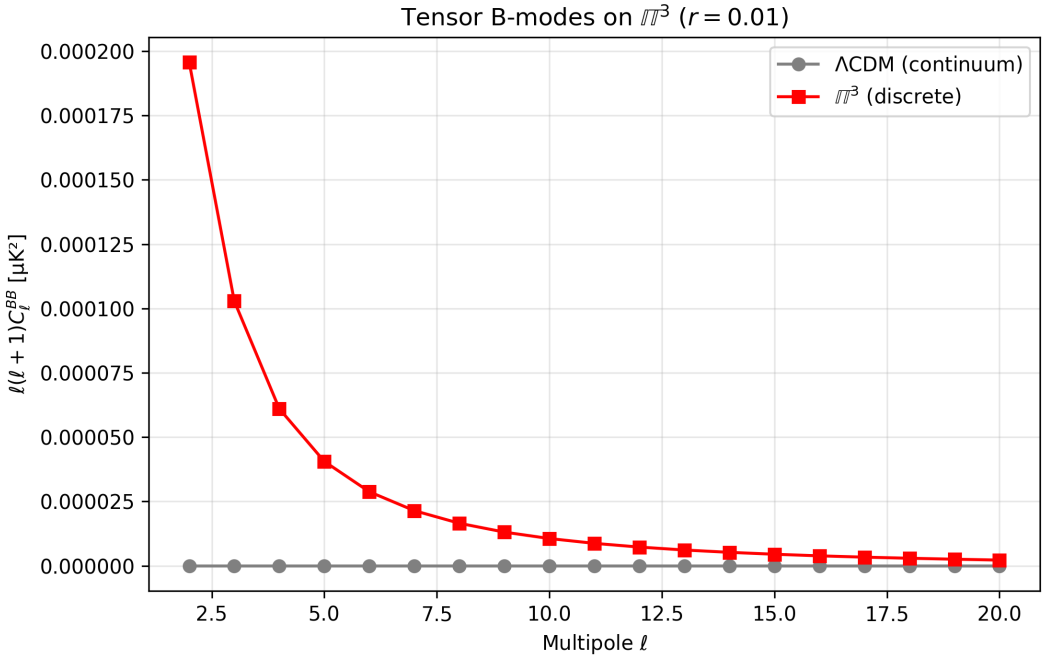

Figure 2: Predicted tensor B-mode oscillations in the IT^3 model (red) compared to ΛCDM continuum (gray). The discrete spectrum induces characteristic modulations at low ℓ.

6 Discussion

The IT^3 model demonstrates that compact flat topology with irrational aspect ratios is:

- **Consistent** with current CMB data (BipoSH PASS, CITS PASS_GEOM)
- **Explanatory** for persistent ΛCDM anomalies without new physics
- **Falsifiable** via clear criteria for upcoming missions (LiteBIRD, CMB-S4, SKA)
- **Reproducible** with fully open code and public Planck data

While not yet statistically preferred over ΛCDM (Bayesian evidence $\ln \mathcal{Z}$ pending), it provides a geometrically motivated, testable alternative worthy of further investigation.

All code, data, and analysis pipelines are publicly available at:

https://github.com/Viktar-Pi/FlatIrrationalTorus

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